Investigating Quantum Hall effect Breakdown in a GaAs-AlGaAs Heterostructure

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Abstract

In this experiment, we investigate the breakdown of the quantum Hall effect (QHE) on a GaAs-AlGaAs heterostructure. We attempt to find estimates of this breakdown current at 300 mBar as well as the behaviour at breakdown and justify this with respect to simple models of the quantum Hall effect. Furthermore, we find the fine structure constant using quantum Hall effect plateaus, and then we attempt to determine the electronic effective mass in our 2D electron gas system using oscillations in resistance that occur due to the Shubnikov-de-Haas effect.

1 Introduction

The classical Hall effect entails the presence of a voltage (often called the Hall voltage) produced across a semiconductor or conductor when a current flows through it with a magnetic field perpendicular to this current. In the quantum Hall effect, we observe that these Hall voltages are quantised, due to the quantised motion of electrons in a magnetic field within Landau levels. This experiment investigates the quantum Hall effect on a 2D-semiconductor heterostructure (a sandwich of GaAs, Gallium Arsenide and AlGaAs, Aluminium Gallium Arsenide), with a particular eye towards investigating the breakdown of the quantum Hall effect, the transition from quantised voltages to the classical, linear Hall effect characteristic. This report begins by examining the theory of the classical and quantum Hall effects, then detailing the experimental setup we used, showing our analyses and some brief qualitative justifications of some of the results we found and thus concluding.
The QHE is interesting for various reasons, first and foremost it was one of the first observed macroscopic quantum phenomena, and has led to further study in the theory of topological condensed matter and the study of exotic particles such as anyons. There have been proposed uses of the quantum Hall effect in quantum computing where a quantum information processor has been defined based on the hyperfine interactions between conduction electrons and nuclear spins in a 2D system in the quantum Hall regime [1]. The results of von-Klitzing's original paper[2] were also used to establish a new standard for the calibration of resistance.[3]

1.1 Classical Hall effect

The classical Hall effect is due to the nature of current in a conductor, and when a magnetic field is present on a conductor, these charges experience a Lorentz force. This can be measured in the form of a voltage V_y perpendicular to the current and applied magnetic field. The measured voltage parallel to the current is V_x. The voltages and currents across the Hall bar (the heterostructure sample) are related via the conductivity/resistivity tensor as shown:

$$V_x = R_{xx}I_x + R_{xy}I_y \tag{1}$$

$$V_y = -R_{xy}I_x + R_{xx}I_y \tag{2}$$

Under classical considerations, such as the Drude model and considering the Lorentz force on electrons due to the magnetic field in this model, we arrive at a model that demonstrates a linear relationship between R_{xy} and magnetic field, and a constant relationship between the magnetic field and R_{xx} via number density as shown: [4]

$$R_{xy} = \frac{B}{n_s e}. \tag{3}$$

This is derived in [4] by applying Newton's equations of motions to a charged electron under crossed electric and magnetic fields, under scattering due to impurities and thermal vibrations with characteristic scattering time

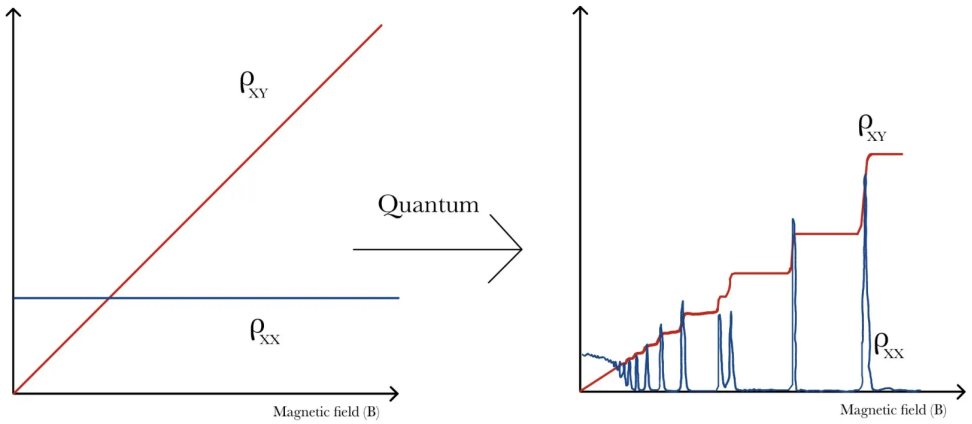


Figure 1: The transition from the classical to quantum Hall effect. The diagram on the left shows how ρ_{xx} and ρ_{xy} vary with magnetic field (line labels depict the relevant y axis). We also see Shubnikov-de-Haas oscillations observed at smaller magnetic fields in ρ_{xx}.

τ. We will find that under very strong magnetic fields and very low temperatures (below a breakdown point in Hall current), this relationship no longer holds, as shown in figure 1.

1.2 QHE in our 2D sample

Electrons in a conducting solid may be treated as free electrons with an effective mass m^* instead of the original electronic mass. This means that the energy E(k) of an electron with momentum $\hbar k$ is given by $E(k) = \frac{\hbar^2 k^2}{2m^*}$. Here we are essentially treating the material structure of our sample as a 2-dimensional electron gas confined an infinite potential well. The energies of these levels have the quantised k-states

$$k_x^2 + k_y^2 = \frac{(2\pi)^2}{L^2}(n_x^2 + n_y^2). \tag{4}$$

Here, n_x and n_y are integers. The number of states N within a circle of radius k is just the area of the circle divided by the area occupied by each individual state which is $\frac{(2\pi)^2}{L_x L_y}$ which gives (when accounting for two electron spins):

$$\frac{N}{A} = 2\frac{\pi k^2}{(2\pi)^2}. \tag{5}$$

In a perpendicular magnetic field, the kinetic energy of the 2D electron system under the assumptions already made becomes quantised into discrete Landau levels representing the quantum harmonic oscillator like so:

$$E_n = \hbar\omega_c(n + \frac{1}{2}), \tag{6}$$

where $\omega_c = \frac{eB}{m^*}$ is the cyclotron frequency under effective mass. This last expression comes from taking the Hamiltonian to be:

$$H = \frac{1}{2m}(\hat{p} - qA)^2, \tag{7}$$

where A is the magnetic vector potential (related to the Aharanov-Bohm effect). Each Landau level has a large degeneracy per unit area, which is given by:

$$\frac{2eB}{h}. \tag{8}$$

This comes from substituting the area of wavevector-space between two states n and n+1 into the formula for the total number of states N within a circle of radius k. This tells us that the degeneracy of each Landau level increases linearly with B while the total electron density remains fixed. Each Landau level can hold at most this number of electrons before beginning to fill the next level and one key difference between our idealised picture of Landau levels as delta functions is that the levels are not perfectly sharp due to the scattering of electrons due to imperfections such as defects at the surface of the layer as well as the addition of impurities. We model this as Landau levels

with a spread of possible energies, i.e. the Landau levels are broadened. This is clearly depicted in figures 2 and 3. The Landau levels are filled upto the Fermi energy E_F, which depends purely on number density in the case of our system. The filling factor gives the number of completely filled Landau levels:

$$\nu = \frac{nh}{eB}. \tag{9}$$

The quantised Hall conductance is then given by:

$$\sigma_{xy} = \nu \frac{e^2}{h}. \tag{10}$$

Here, conductivity is inverse to resistivity. This formula in its exact form comes from interesting topological considerations [5] - specifically related to the TKNN formula and Chern numbers. The TKNN formula is in fact a statement that the Hall conductivity is a topological invariant of the system. Via a simple model, it can be thought to be because for strong magnetic fields and Fermi energies between Landau levels, the bulk becomes insulating , edge states or conducting channels that exist at the boundaries of a 2D electron gas contribute the most to conductivity and each filled level contributes e^2/h. That said, there is far more topological nuance to this that can be found in [5]. The fomrula (10) means as magnetic field is changed, the number of completely filled Landau levels may not change which directly gives rise to a plateau, and therefore the quantum Hall effect.

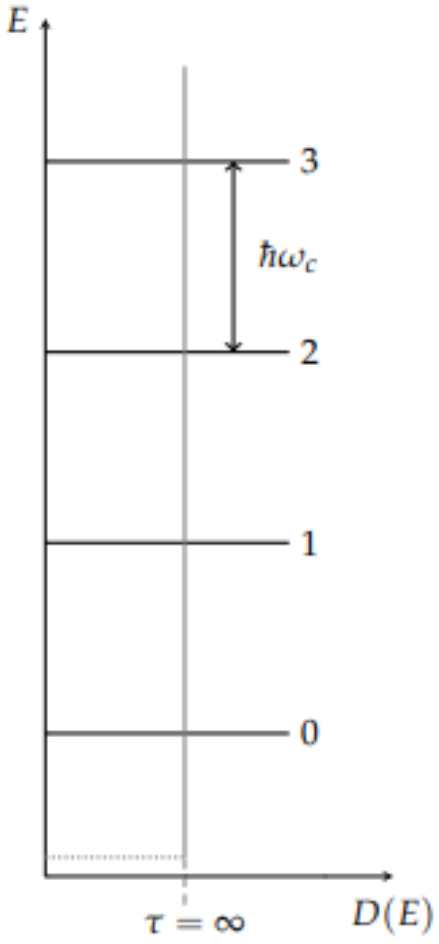


Figure 2: Discrete Landau levels, prior to broadening due to scattering. From the SS07 lab script.[4]

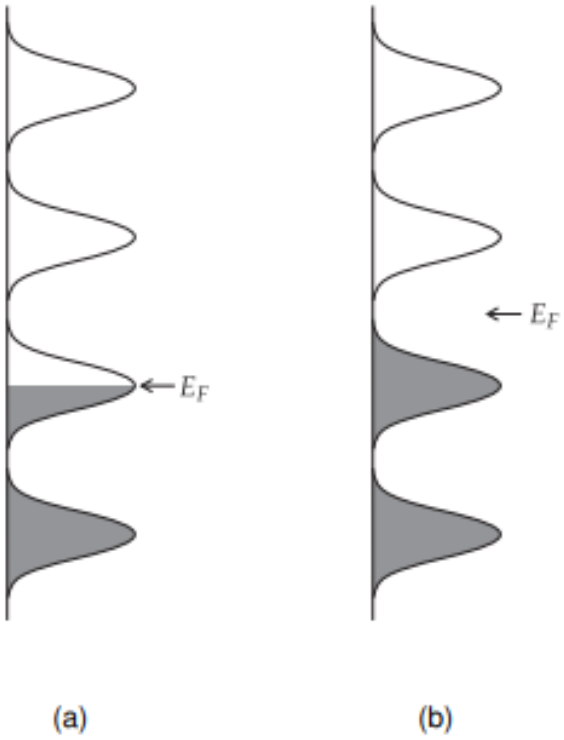


Figure 3: Broadened Landau levels, after broadening due to scattering. From the SS07 lab script.[4]

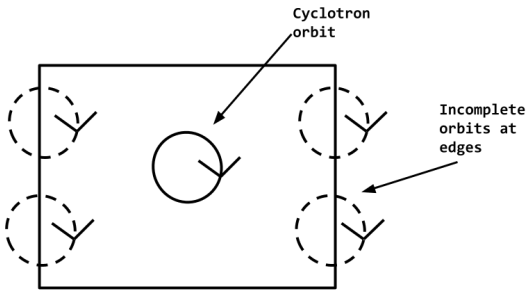


Figure 4: Diagram that demonstrates scattering at the edges of a sample, thus leading to the transition from Shubnikov-de-Haas oscillations to spikes in ρ_{xx}.

1.3 The use of GaAs-AlGaAs heterostructures to form a 2DEG

The reason we use a GaAs-AlGaAs heterostructure in this experiment in order to observe the quantum Hall effect and related phenomena, is because they produce an extremely high-mobility 2D electron gas (2DEG). GaAs and AlGaAs have very similar lattice constants and compatible band structures. Due to very small lattice mismatches, there are very few defects at the interface, reducing electron scattering and allowing electrons to travel long distances without losing momentum. 2D electron gases are an ideal candidate to observe QHE effects because electron motion perpendicular to plane is frozen, and furthermore the 3-dimensional variant of this requires a theoretical extension of the single-electron Landau quantisation framework as developed for 2D electron systems, as discussed in [6].

1.4 Shubnikov-de-Haas oscillations

The Shubnikov-de-Haas (SdH) oscillations are oscillations of the resistivity parallel to the current in a 2D electron gas. As mentioned earlier, we have slightly broadened Landau levels due to disorder, hence as B increases the Fermi level periodically crosses Landau levels. Each crossing causes a peak in R_{xx} leading to oscillations that are periodic in $\frac{1}{B}$.This is because the filling factor is $\nu = \frac{nh}{eB}$ and oscillations occur whenever the filling factor passes through an integer $\nu = 1, 2, 3, ...$ We may rearrange this to obtain that $\frac{1}{B} = \frac{\nu e}{nh}$. Furthermore, as shown in figure 4, although electrons in the bulk of the sample are able to undergo full cyclotron orbits, electrons at the edges of the sample are backscattered. This results in so called edge states. We find that as we raise the magnetic field to several Tesla, the minima of the oscillation begin reaching zero due to edge states, as the bulk is eventually completely depleted of conducting electrons.[7] We use that the amplitude of Shubnikov-de-Haas oscillations varied with B is dependent on the cyclotron frequency, which is then dependent on the electronic effective mass. We can therefore use a fit on this to determine the effective electronic mass. Details are explained in section 3.1.

1.5 Quantum Hall effect breakdown

This subsection briefly introduces some of the theory behind the breakdown of the Hall effect, which we build up in more detail in the analysis in order to explain some of the results that we have obtained. Under the quantum Hall effect, the longitudinal resistivity vanishes (other than spikes) and the Hall resistivity ρ_{xy} is quantised. This quantisation remains robust over a range of magnetic fields, forming a plateau. This begins to break down as the current through the sample exceeds a critical value I_c. A current flowing through the Hall bar produces a transverse Hall electric field. The field is perpendicular to the current and to the magnetic field. As I increases, the transverse field grows proportionally. As this field becomes large, the Landau levels become tilted in energy. An electron sitting at position x across the width of the bar has its Landau level energy shifted by eE_Hx where E_H is the transverse electric field. The tilt means that adjacent Landau levels (normally separated by $\hbar\omega_c$can cross each other spatially at some point across the width of the sample. When the tilting due to the transverse field is steep enough that the energy gained by an electron drifting a distance of order $l_B = \sqrt{\hbar/eB}$ (the magnetic length, the natural length scale of quantum mechanics in a magnetic field which comes from the wavefunction scale set by the Hamiltonian in (6)) is on the order of the Landau level transition energy $\hbar\omega_c$, inter Landau level transitions become possible. Below a critical current, electrons remain confined to their Landau levels and the quantisation is preserved.[8] Above it, electrons may scatter between levels, leading to the breakdown of the quantisation. There are various competing models and semiclassical theories that attempt to understand the microscopic mechanism

by which electrons scatter between Landau levels. One such example is the QUILLS model [8] (which we discuss further in the context of our results). This model considers that electrons moving through the bulk of the 2DEG can scatter off impurities, defects and other inhomogeneities, thus hopping some distance Δx in the Hall field direction. What we require for inter Landau level scattering is that the hops lead to an energy gain on the order of $\hbar\omega_c$. For higher transverse electric fields, this hopping distance becomes smaller, and once it becomes comparable to the mean free path for impurities, inter-Landau-level transitions become probable, leading to breakdown. This will be discussed further in section 3.4. Competing models include the electron heating model[9], which proposes that once a small population of electrons is excited to higher Landau levels by the Hall field, the hot electrons can themselves generate further scattering through electron-electron interactions. The phonon emission model [10] considers that electrons accelerated by the Hall field can emit optical or acoustic phonons to relax between Landau levels, particularly at higher temperatures. In practice, for our sample, the effects are most dominated by QUILLS.

2 Experimental setup

2.1 The Hall bar sample

The sample we use to investigate the QHE is 2-dimensional electron system confined in a GaAs-AlGaAs heterojunction [4] - this is produced by growing a thin (10nm) layer of material on a crystalline substrate. For quality purposes, this is using two materials with near identical lattice constants - as we would otherwise be introducing defects at the interface. This is why we use GaAs and AlGaAs, as discussed earlier in section 1.3. The sample has been lithographically defined and etched into the shape of a Hall bar, in order to reduce the effects of contact resistances and further enable us to make measurements of Hall voltage and magnetoresistance.

The sample is mounted on a probe and kept in thermal contact with a liquid ^{4}He bath. To reach lower temperatures, the vapour pressure above the liquid is reduced using a pumping system. Lowering the pressure causes the Helium to boil, removing latent heat and cooling the liquid until it reaches the boiling point corresponding to the new lower vapour pressure. The pump-down from 4.2 K to 1.6 K is performed over atleast thirty minutes to avoid violent boiling that would waste Helium.

2.2 The superconducting magnet

The magnet consists of two coils made of NbTi and Nb_3Sn, both of which are superconducting at liquid Helium temperatures. The use of a superconducting magnet is required in order to produce sustained large magnetic fields (up to a maximum limit of 6.87 T). Furthermore, the QHE requires that the cyclotron frequency ω_c must greatly exceed the scattering rate. This remands fields of several Tesla for the rough effective mass of the electron in GaAs. This requires the use of a superconducting magnet.

The magnetic flux density is controlled and swept via computer software, which allows for slow continuous sweeps between field values. This sweep capability allows us to observe SdH oscillations in V_{xx} at low fields below 1 T as well as QHE plateaus at higher field values. We must keep sweep rate slow to avoid eddy current heating of the sample and to ensure that the system remains in thermal equilibrium throughout the measurement. [4] A critical experimental and safety consideration with regard to the superconducting magnet is the risk of a quench, where the magnet suddenly loses its superconducting state and becomes resistive. This dissipates the stored inductive energy nearly instantaneously, causing violent boiling of several litres of ^4He and producing several cubic litres of gas. To prevent quenches, the power supply must not be switched off abruptly and the magnetic sysstem must not be perturbed. When measurements were paused, we ramped the field down to 0 T to reduce wastage of helium.

2.3 Data logging and collection

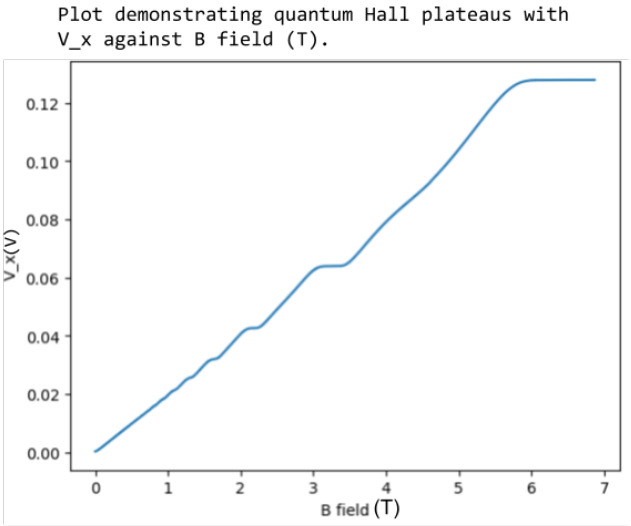
Data is read from three Keithley 2000 DMMs representing V_x, V_y and V. We have the following relations that allow us to consider the variance of resistance through and across the sample.

$$R_{xx} = V_x/I, R_{xy} = V_y/I. \tag{11}$$

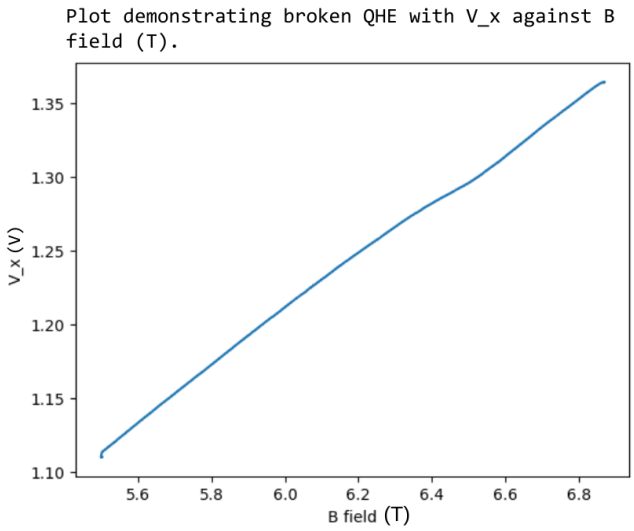
2.4 Measurement

The general course of measurement during the experiment is as follows: Sweep magnetic field over sample and measure V_X, V_Y over the course of the sweep. We occasionally localised the magnetic field over a certain region if

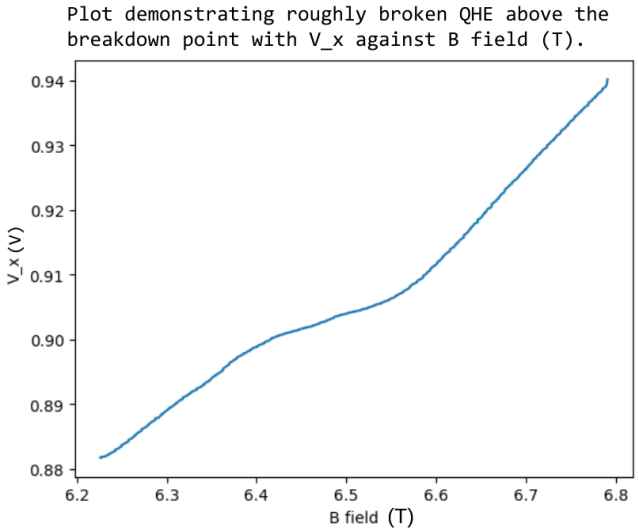
we wished to study that region in more detail (for example, earlier on the in the experiment). When considering SdH oscillations we kept the magnetic fields lower, as lower B-fields are where the SdH oscillations are observed.



(a) quantum Hall effect plateaus observed at 10 µA and approximately 370mBar.



(b) Broken quantum Hall effect showing classical Hall-effect characteristics at 100 µA and 270mBar.



(c) Broken quantum Hall effect at 70 µA and 270mBar, approximately 10 µA above the breakdown current at this pressure.

Figure 5: Comparison of the intact and broken quantum Hall effect regimes under different current and pressure conditions.

3 Analysis

3.1 Determination of electronic effective mass

Firstly, we find the amplitudes of Shubnikov-de-Haas oscillations by finding the difference between subsequent maxima and minima in V_x. This is done by numerically finding turning points in the graph of SdH oscillations, i.e. by finding points $[y_{i-1}, y_i, y_{i+1}]$ where $y_i > y_{i-1}, y_{i+1}$, a maxima or $y_i < y_{i-1}, y_{i+1}$, a minima. We may find

amplitudes by comparing the differences between maxima and minima. This can be seen as an example in figure 6.

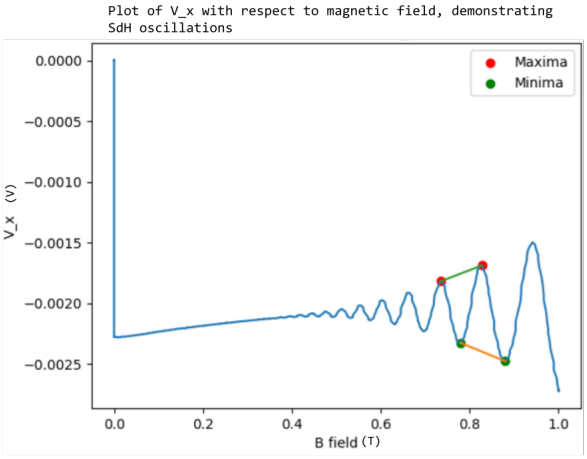


Figure 6: Finding SdH oscillation amplitudes at 0.8 T

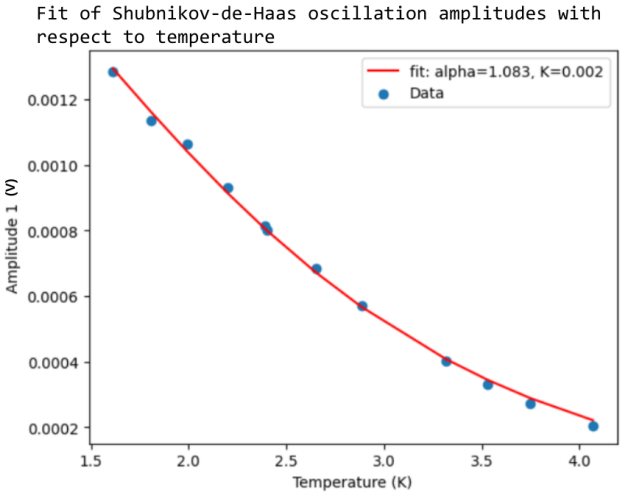


Figure 7: Fitting Shubnikov-de-Haas oscillation amplitude to temperature

We may use the amplitudes of Shubnikov-de-Haas oscillations fit to temperature via:

$$A(T,B) = K(B)\frac{\chi}{\sinh(\chi)}. \tag{12}$$

where $K(B)$ is a function of the magnetic field only and $\chi = 2\pi^2 \frac{k_B T}{\hbar\omega_c}$. This is an approximation to the Lifshitz-Kosevich formula, originally derived for three-dimensional metals [11] and describes how quantum oscillations (periodic variations in magnetisation and resistivity as a function of $\frac{1}{B}$) arise from Landau quantisation of the electron spectrum. The formula encapsulated by $K(B)$ is known as the Dingle factor. This describes damping due to disorder, and is characterised by the quantum scattering time. In the experiment to determine Shubnikov-de-Haas oscillations, since we are fixing B and varying T, we find that $K(B)$ is a constant. We may then fit to this function using scipy, with uncertainties estimated from covariance matrices with this fit.

This gives the following fit shown in figure 7. The fit yields an electronic effective mass of $m^* = (5.2 \pm 0.1) \times 10^{-32}$kg.

3.2 Fine structure constant determination

We measure the longitudinal and transverse magnetoresistance to derive the fine structure constant to a high precision of one part in 10^5.
For a truly two-dimensional system, the width of the sample cancels out so that [4]:

$$\rho_{xy} = \frac{|V_x|}{I} = \frac{1}{j}\frac{h}{2e^2}. \tag{13}$$

The value of the resistance determined only by the fundamental constants h and e, is incredibly accurate. Although [4] calculates it only for a completely filled Landau level, it remains constant throughout the plateau. We may rewrite this in terms of the fine structure constant as such:

$$\rho_{xy} = \frac{1}{j}\frac{\mu_0 c}{4\alpha}. \tag{14}$$

What we may then do is choose a number of Hall plateaux and use them to make a measurement of the resistance quantum $\frac{h}{2e^2}$ and thus the fine structure constant. The digital multimeters remain uncalibrated, but have high precision. The current source has low precision and accuracy, while the standard resistor is calibrated very well to a resistance of 10000.90Ω.
Hence, after varying our magnetic field, we found a plateau at $6.30 \pm 0.005T$ which is at $j = 1$ (found upon

consideration of carrier density). Having taken a measurement of transverse voltage and current I, and using the equation above, we arrive at $\alpha = 7.30 \pm 0.01 \times 10^{-3}$. This is consistent with the accepted value of $\alpha = 7.2973525643 \pm 0.0000000001 \times 10^{-3}$ from [12].

3.3 Quantifying quantum Hall effect breakdown

The aim of our experiment is to attempt to find and investigate the point of breakdown of the quantum Hall effect, particularly considering variations in current. Under the quantum Hall effect, sweeping the Magnetic field should produce plateaus in V_y. This can be seen in the plots in figure 5. As may be seen, the minimum gradient appears to go above zero beyond the breakdown point, which is something we will consider. In plot 5a), the QHE is entirely intact while it breaks in 5c) and further reaches the classical Hall effect characteristic in 5c). Quantum Hall plateaus arise because disorder localizes states between Landau levels, preventing changes in the number of current-carrying extended states as magnetic field varies; the plateaus break down when the Fermi energy crosses extended states or when high current, temperature, or reduced magnetic field enables inter-Landau-level scattering. This gives us a possibility for a quantitative notion of quantum Hall effect breakdown - essentially the point at which we stop seeing plateaus upon a sweep with the magnetic field. We determine if our sweep of the magnetic field contains plateaus by considering the **minimum absolute discrete gradient**. This is found by generating an array of $[x_{i+1} - x_i]\ \forall i$ and considering the minimum value in this array. Having done this, we may vary the current through the Hall bar and observe changes in the minimum gradient, producing a plot as shown in figure 10. This plot keeps the quantum number constant, however involves finding minimum gradients at a variety of different temperatures/pressures. An additional plot is given in figure 11, which keeps the pressure (and therefore temperature) constant.

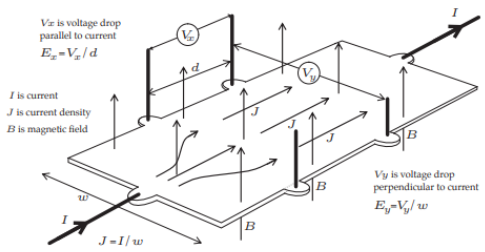


Figure 8: Geometry of the Hall bar sample, from the SS07 lab script.[4]

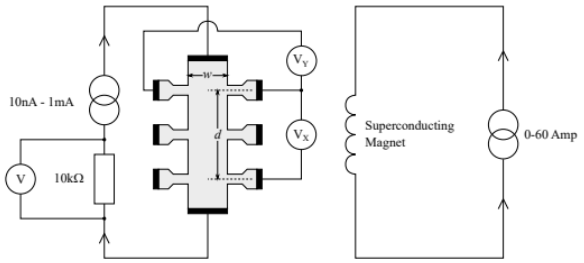


Figure 9: Sample geometry and wiring diagram, from the SS07 lab script.[4]

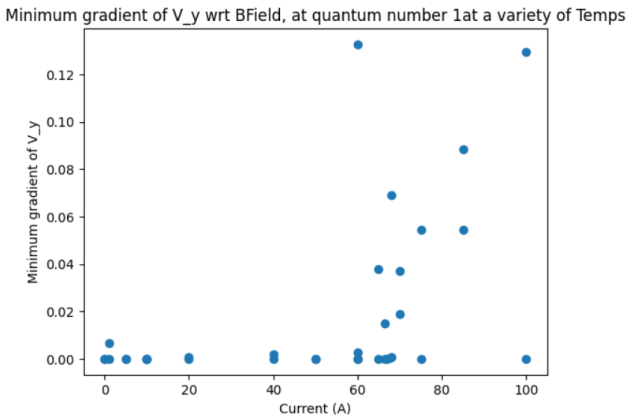


Figure 10: Minimum gradient of V_y with respect to magnetic field at quantum number 1 at a variety of temperatures.

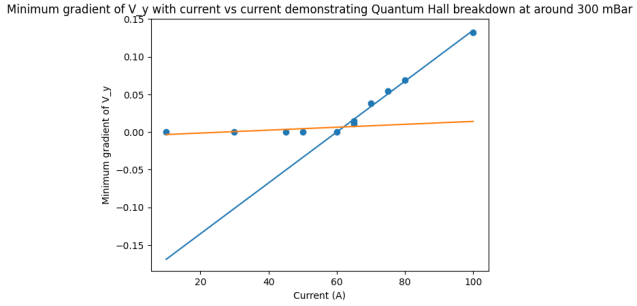


Figure 11: Demonstration of the breakdown point algorithm for QHE breakdown at 300 mBar. The breakdown current can be found to be 62 µA. The two lines represent residual-minimising fits of two segments of the data. Their intersection point gives a rough estimate of the breakdown point.

3.4 An algorithm to find the breakdown point

As can be observed in figure 10, the transition from quantum Hall effect to classical Hall effect is rather sharp - we go from having near 0 minimum gradient across a range of current to having a roughly linear relationship. One way to estimate a rough breakdown current is to draw two separate lines of best fit, one within the QHE regime and one within the classical Hall effect regime and finding their point of intersection. Finding these lines of best fit can be done by attempting to partition consecutive chunks of the minimum gradient data into [QHE regime; Classical regime] and separately applying linear regression fits to the QHE regime data and classical regime data until we find a partition and lines of best fit that minimise residuals. We then find a point of intersection and refer to that as our point of Hall effect breakdown. A key point is that this model of quantum Hall effect breakdown makes the assumption that the minimum gradient after breakdown varies linearly with applied current on the sample, which is evidenced by the data shown in figures 4 and 5. We will briefly try to explain this linearity in the next subsection. From our algorithm, applied especially to the data at 300mBar, we find a breakdown point of 62 µA. This data is shown in figure 11 with the plotted lines.

3.5 Minimum gradient linearity

In this section, we attempt to give a qualitative justification of the linearity in minimum gradient that we experimentally discovered with reference to a documented model.

In their 1986 paper, Eaves and Sheard introduced a semiclassical model for the QHE known as quasi-elastic inter-Landau-level scattering (QUILLS), which we discussed briefly when considering the theory behind QHE breakdown. Expanding on our previous Landau level based explanation of the quantum Hall effect; electrons move in circular orbits and their energies are grouped into Landau levels. At low currents, electrons stay within their Landau levels and the Hall resistivity ρ_{xy} is flat. The breakdown of the quantum Hall effect happens when we reach a critical breakdown current. Past this current, the strength of the transverse electric field (the Hall electric field) becomes strong enough that electrons can jump between Landau levels, breaking the quantisation.[2] QUILLS explains the mechanism by which electrons may jump between Landau levels past this critical breakdown current, and subsequently explains the linearity of the minimum gradient. It explains that electrons can scatter between Landau levels without changing much energy - via both the strong transverse field and quasi-elastic scattering off disorder and impurities in the material. As we move above the critical current, electrons scatter more frequently between levels as the rate of electron scattering is predicted (by QUILLS) to increase with electric field. As the rate of electron scattering increases, the deviation of ρ_{xy} from a flat gradient in Landau levels with respect to the magnetic field increases. This results in a roughly linearly increasing gradient with current in the inter-Landau level regions - thus giving rise to a roughly linear increase in minimum slope of ρ_{xy}.

More concretely explaining the linearity, to move between Landau levels, an electron must gain $\hbar\omega_c$ in energy. An electron moving a distance Δx along the transverse Hall field gains energy:

$$\Delta\epsilon = eE\Delta x. \tag{15}$$

From our required energy to move between Landau levels, we arrive at an approximate required spatial hop of:

$$\Delta x \approx \frac{\hbar\omega_c}{eE}. \tag{16}$$

Scattering against impurities allows electrons to hop sideways by a typical distance (the mean free path of the scattering, for example), to first order this changes very little. We can consider the hop distance to be distributed around this typical distance. As electric field, E increases, the required hop distance decreases, thus increasing the probability of a successful hop with E. In the QUILLS model, we assume the distribution of possible impurity-induced displacements is roughly smooth (an assumption justified in earlier work on hopping theory, especially pioneered by Efros and Shklovskii and described in depth in [13]). To first order, this results in a linear relationship between electric field E and scattering rate, and furthermore the scattering rate is proportional to $\frac{d\rho_{xy}}{dB}$ in the quantised Landau level regions. This qualitatively justifies the linearity of minimum gradient. The QUILLS model is a relatively simple model, and more complicated models account for competing effects.

3.6 Sources of noise and uncertainty

There are various sources of error within the experiment that must be acknowledged, with some attempt at mitigation. The first is thermal or Johnson-Nyquist noise, the random thermal motion of electrons in wires and contacts

which produces voltage fluctuations. We may attempt to mitigate these by using cryogenic temperatures. Another potential issue is small fluctuations in the current of the superconducting magnet, which can lead to uncertainty in B, an assumed variable. This may also lead to uncertainty in plateau positions, and propagate to our discussion of breakdown points. Our calculation of the electronic effective mass from SdH amplitudes is also affected by the discrete sampling resolution, leading to amplitudes that may be smaller than they really are (as the real maximum may not have been sampled). Poor electrical contacts to the Hall bar may also introduce fluctuating voltages. Another major issue encountered while attempting to find the breakdown current is that it is very difficult to keep the pressure and temperature constant, hence while we can computationally filter for quantum number using the Landau levels, there generally may be a variety of temperatures.

4 Conclusion

We find from Shubnikov-de-Haas oscillations an electronic effective mass of $m^* = (5.2 \pm 0.1) \times 10^{-32}$kg within our 2D electron gas. We also use a quantum Hall plateau to arrive at a value of the fine structure constant, $\alpha = 7.30 \pm 0.01 \times 10^{-3}$. Furthermore, we clearly observe quantum Hall effect breakdown at a particular current of 62 µA at 300mBar. We also observe that the minimum gradient seems to vary very linearly with current past the breakdown point, which we have qualitatively justified using the quasi-elastic inter-Landau level scattering theory.

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